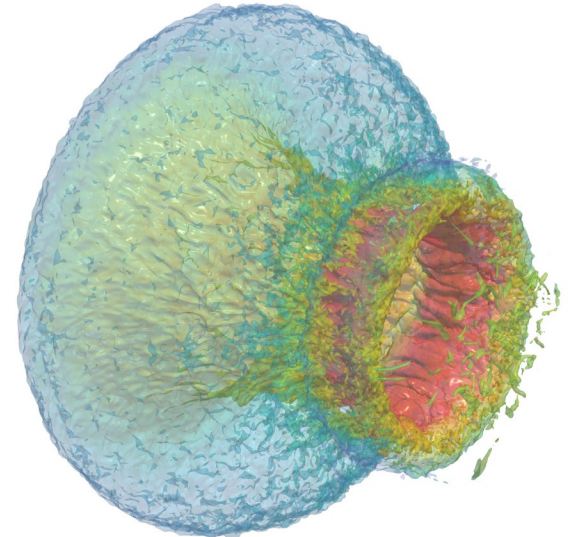
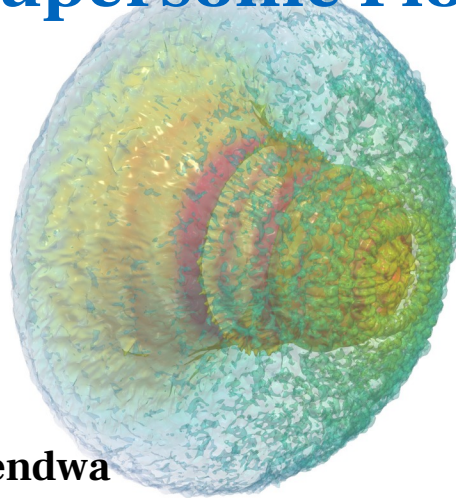
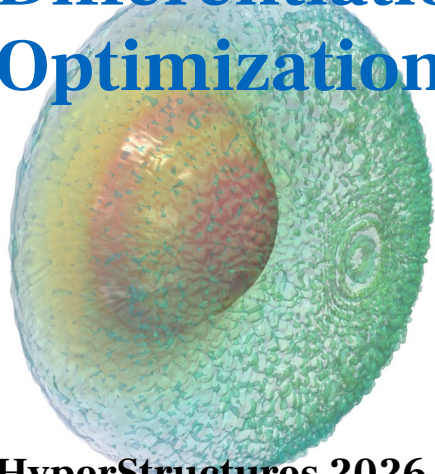


# JAX-Fluids Part 2: Leveraging Automatic Differentiation for Shape and Material Optimization in Supersonic Flows



**HyperStructures 2026**

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# Overview Optimization Cases

**Data assimilation:** Reconstruct flow field and shape from sparse data measurements

$$M = 2, Re = \infty$$

Density

**Shape optimization:** Minimize drag while maintaining initial volume

$$M = 2, Re = 300$$

Pressure

**Material Optimization:** Minimize solid temperature with given amount of insulator material

$$M = 2, Re = 300$$

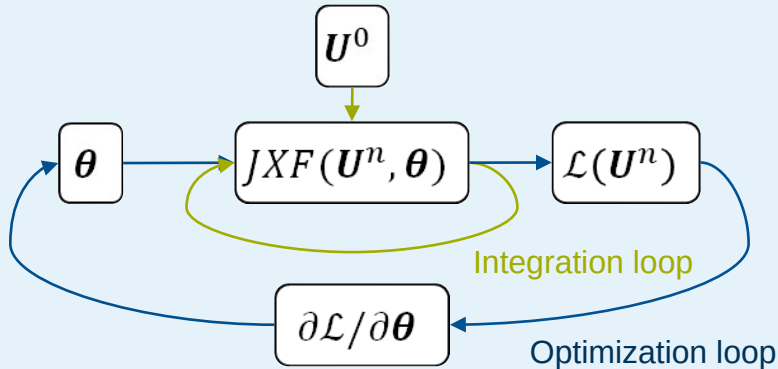
Mask

Temperature

# Optimization Paradigms: Unrolling vs. ODIL

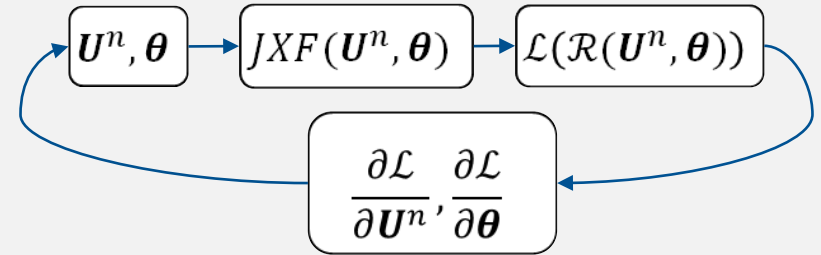
## Unrolling the solver in time

- Parameters  $\theta$  define shape or material properties
- Given an initial condition  $U^0$ , JAX-Fluids (JXF) is unrolled in time, producing a flow field trajectory  $U^n$



## Optimizing a Discrete Loss (ODIL)

- Parameters are the discrete flow field  $U^n$  plus the shape parameters  $\theta$
- Discrete PDE residual  $\mathcal{R}(U^n, \theta)$  is imposed directly in loss function  $\mathcal{L}$



# Shape representation: Parametric vs. Free-form

## Parametric

- Small set of parameters  $\theta$  define shape
- Analytical representation of the shape, e.g., cubic splines
- Level-set  $\phi$  is computed numerically from parameters  $\phi = f(\theta)$
- When computing  $\phi = f(\theta)$  we explicitly enforce signed distance property  $|\nabla\phi| = 1$

## Free-form

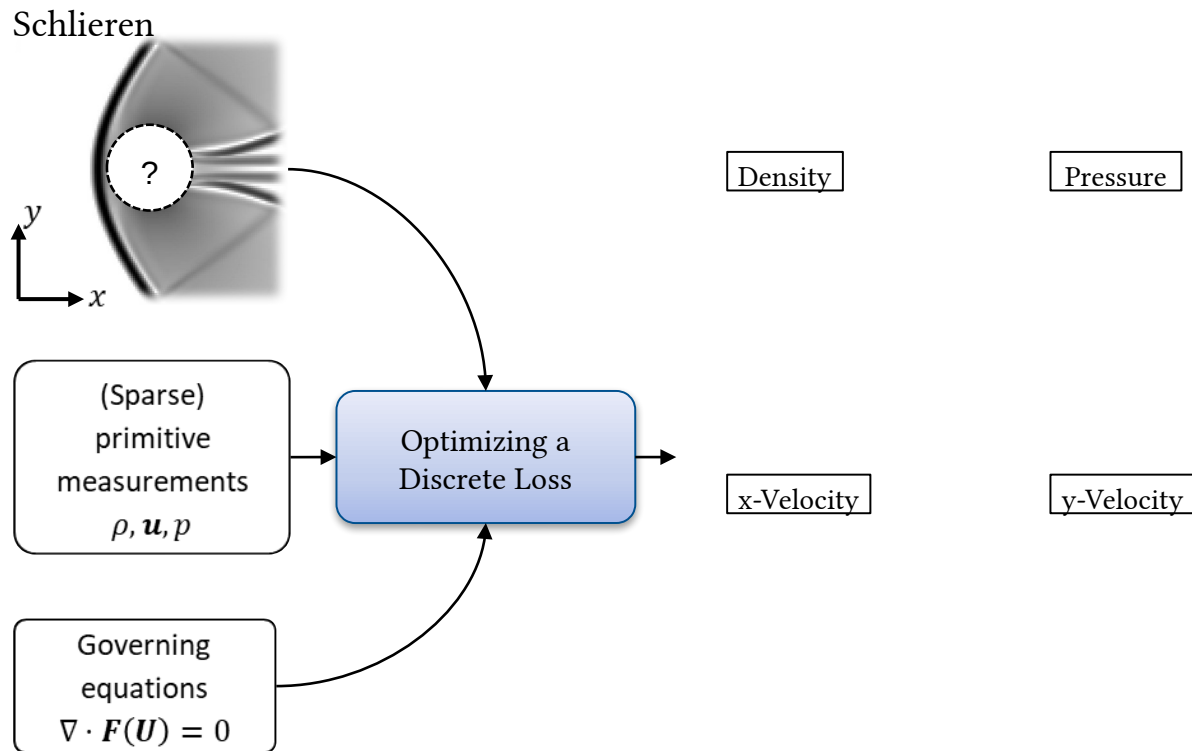
- Set of parameters  $\theta$  are the values of the discrete level-set field  $\phi$
- Signed distance property must be maintained
  - Soft penalty in loss function
  - Solving  $\frac{\partial\phi}{\partial t} + (|\nabla\phi| - 1) = 0$  explicitly after each optimization step

# **Data Assimilation to Reconstruct Flow Field and Shape**

# Data Assimilation to Reconstruct Flow Field and Shape

Mach 2 flow around a cylinder

Assuming **unknown outer boundary conditions and solid shape**, the flow field and solid shape is inferred from schlieren visualizations and sparse measurements of primitive variables



# Data Assimilation to Reconstruct Flow Field and Shape



- $\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{Data} + \mathcal{L}_{Reg}$
- **Parametric** shape representation: Learning the radius  $R$  and compute level-set analytically  $\phi = f(R)$
- **Free form** shape representation: Learning the discrete level-set field directly

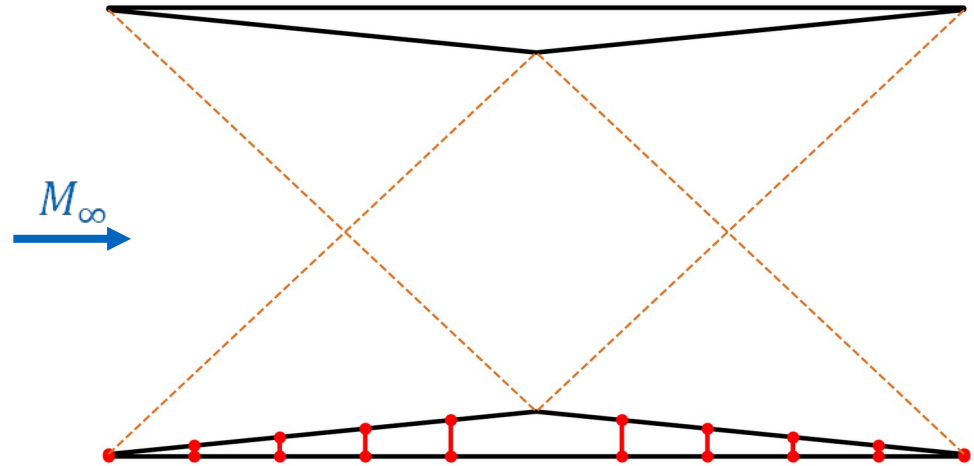
# Shape Optimization



# Parametric Shape Optimization

- Busemann biplane minimizes wave drag at design Mach number
- How do we have to adjust the shape to minimize wave drag across a number of Mach numbers?

- $\mathcal{L} = \sum_{i=1}^N c_d(Ma_i)$



# Parametric Shape Optimization



# Free-form Shape Optimization

- What is the shape that minimizes drag while maintaining a reference volume?
- Parameters  $\boldsymbol{\theta}$  are the discrete level-set values  $\boldsymbol{\theta} = \phi$
- Loss function  $\mathcal{L}(\boldsymbol{\theta}) = C_d + (V - V_{ref})^2$
- Algorithm for an optimization step:
  1. Compute gradient  $g = \partial \mathcal{L} / \partial \boldsymbol{\theta}$
  2. Update the level-set by solving  $\frac{\partial \phi}{\partial t} + \mathbf{u}_\Gamma \cdot \nabla \phi = 0$ , with  $\mathbf{u}_\Gamma = \mathbf{n}_\Gamma \cdot g$
  3. Reinitialize the level-set by solving  $\frac{\partial \phi}{\partial t} + (|\nabla \phi| - 1) = 0$

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# Material Optimization

# Composite Material Optimization

- How do we design a composite consisting of materials 1 and 2, with  $\lambda_1 \ll \lambda_2$ , to minimize the shock-induced temperature increase within the solid?

Cylinder

Sphere

# Composite Material Optimization

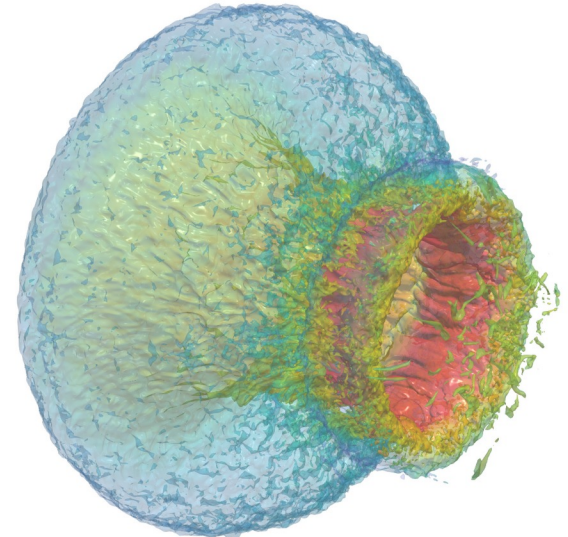
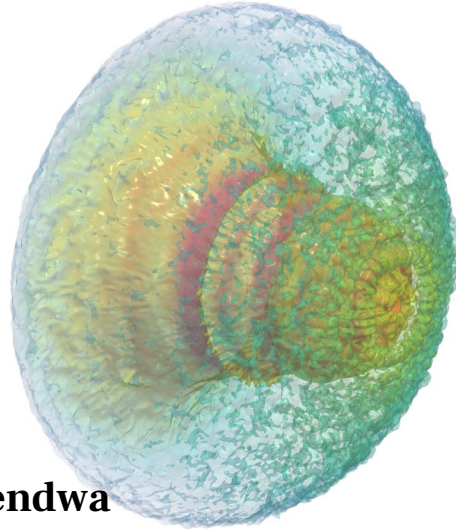
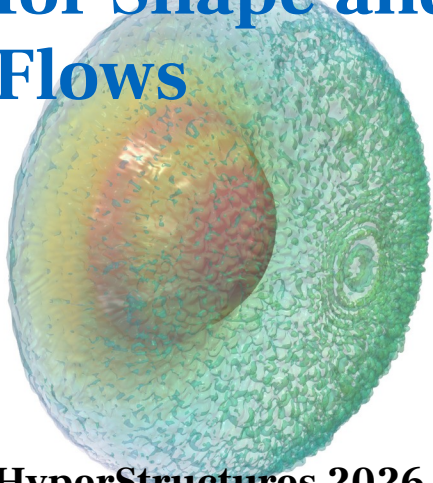
- Learn scalar mask  $M \in [0,1]$  representing cell-wise volume fraction of the insulator
- **Composite heat conductivity:**  $\lambda = M\lambda_1 + (1 - M)\lambda_2$  with  $\lambda_1 \ll \lambda_2$
- $\mathcal{L} = \bar{T}_s + (\bar{M} - \bar{M}_{ref})^2$  with  $\bar{M}_{ref} = 0.2$



# Conclusion

- Showcased differentiability on simple problems (laminar, low Mach)
- Goal: Extend to industry scale problems (WMLES)
- **Long temporal trajectories are challenging:**
  - Memory consumption during backward pass
  - Stability of gradients, especially for nonlinear numerical methods

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